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2. Eigenvalues and eigenvectors in MATLAB

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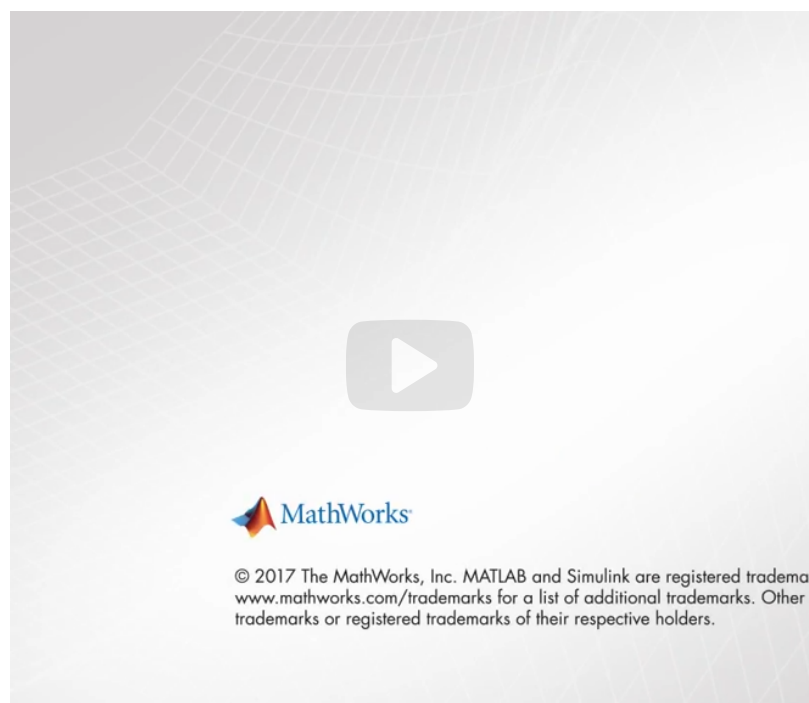
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Calculator

Recitation due Feb 14, 2024 20:30 IST

Finding eigenvalues and eigenvectors in MATLAB

Start of transcript. Skip to the end.



Many simulation and data processing tasks require calculating eigenvalues and eigenvectors

of a matrix.

Car designers need them to damp out road noise, while a biologist may use them to study population dynamics.

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Finding eigenvalues and eigenvectors using MATLAB

If **A** is a square matrix, then we write the following command in MATLAB

```
[V,D] = eig(A)
```

This generates two square matrices:

1. $\mathbf{V}$ is a 3×3 matrix, where each column is a nonzero eigenvector of $\mathbf{A}$.
2. $\mathbf{D}$ is a diagonal 3×3 matrix. The elements along the diagonal are the eigenvalues of $\mathbf{A}$.

Write a MATLAB script which calculates the three eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ of the matrix A and the corresponding eigenvalues $\lambda_1, \lambda_2, \lambda_3$. Express the eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in terms of the unit vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, where

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 5 & 1 \\ 1 & 0 & 3 \end{bmatrix}.$$

Script ?

Save Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 % Matrix A is provided for you.
2 A = [1 -1 0; 2 5 1; 1 0 3];
3 % Now use eig(A) to find the eigenvalues and eigenvectors of A
```

 Calculator

```
4 [V,D] = eig(A);
5 % Now extract the three eigenvectors of A
6 % and define them as three separate column vectors v1, v2, v3
7 v1 = V(:,1);
8 v2 = V(:,2);
9 v3 = V(:,3);
10 % Now extract the three correspondong eigenvalues of A
11 % and define them as three separate variables e1, e2, e3.
12 % These should be numbered so that A*v1 = e1*v1, etc.
13 e1 = D(1,1);
14 e2 = D(2,2);
15 e3 = D(3,3);
```

▶ Run Script

?

Assessment: All Tests Passed

Submit

?

- ✔ Eig used
- ✔ Check v1, v2, and v3 exist, are nonzero, and are distinct
- ✔ Check eigenvectors and eigenvalues correctly defined

Output

Code ran without output.

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2. Eigenvalues and eigenvectors in MATLAB

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